Web Application Overview: Cochlear Explorer

The **Cochlear Explorer** is an interactive, web-based 3D visualization tool designed to provide a comprehensive understanding of cochlear implant systems. It serves as both an educational platform and a technical guide for users, medical professionals, and technicians.

Key Features and Functionality

- **Interactive 3D Model:** The core of the application features a high-fidelity 3D model of a cochlear implant. Users can rotate, zoom, and inspect the device from every angle to understand its physical design and ergonomics.
- **Component Hierarchy & Exploration:** A detailed sidebar allows users to navigate through the "Nucleus hierarchy." This includes:
    - **Main Body:** Exploration of the processor unit.
    - **Mechanism:** Detailed views of microphones, control buttons, and indicator lights.
    - **Connector & Cable:** Visualization of the slimline cables and magnets that connect the external processor to the internal implant.
- **Customization Settings:** The "Model Settings" menu allows users to visualize different configurations, such as:
    - **Color Selection:** Changing the device color (e.g., Beige, Black, etc.) to see how it looks in real life.
    - **Battery Types:** Swapping between various battery sizes (e.g., Standard vs. Long) to understand how they affect the device's footprint.
- **Integrated Documentation:** For every selected part, the app provides instant access to essential technical resources:
    - **Schematics:** Technical drawings for engineering or repair insights.
    - **Manuals:** Step-by-step user guides.
    - **Maintenance:** Instructions on how to care for and clean the specific component.

What Does the App Do?

The application bridges the gap between complex medical technology and user accessibility. By using the **Cochlear Explorer**, users can virtually "disassemble" the device to see how it works, learn how to maintain specific parts, and customize the hardware to fit their needs before ever handling the physical device.

It is the ultimate digital twin for cochlear technology, ensuring that information is always just a click away.